Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #16

Remote Patient Vitals Alert & Monitoring System

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Problem Overview

A **Remote Patient Vital Alert & Monitoring System** that continuously collects patient vital signs such as **SpO2, heart rate, and blood pressure**, detects abnormal readings using predefined clinical thresholds, sends real-time alerts, and automatically escalates critical situations to the appropriate caregivers.

Actors:

- Patient
- Caregiver / Healthcare Provider
- System Administrator
- **Architectural Style Analysis**

Architecture	Advantages	Disadvantages	Suitability
Layered	Clear separation of concerns, easy maintenance, simple development and testing	Some dependency between layers	**High**
Microservices	Independent scaling, fault isolation, suitable for handling multiple patient monitoring services	More complexity and network overhead	**High**
Client-Server	Simple and centralized, easy to implement and manage	Scalability issues and potential single point of failure	**Medium**

Selected Architecture: Layered Architecture

Reason: The application has clearly separated responsibilities such as **patient data collection, vital-sign processing, threshold breach detection, clinical alerting, emergency escalation, and data storage**. Layered architecture provides a simple and maintainable structure for these functions, while making the system easier to develop, test, and maintain.

Component Identification

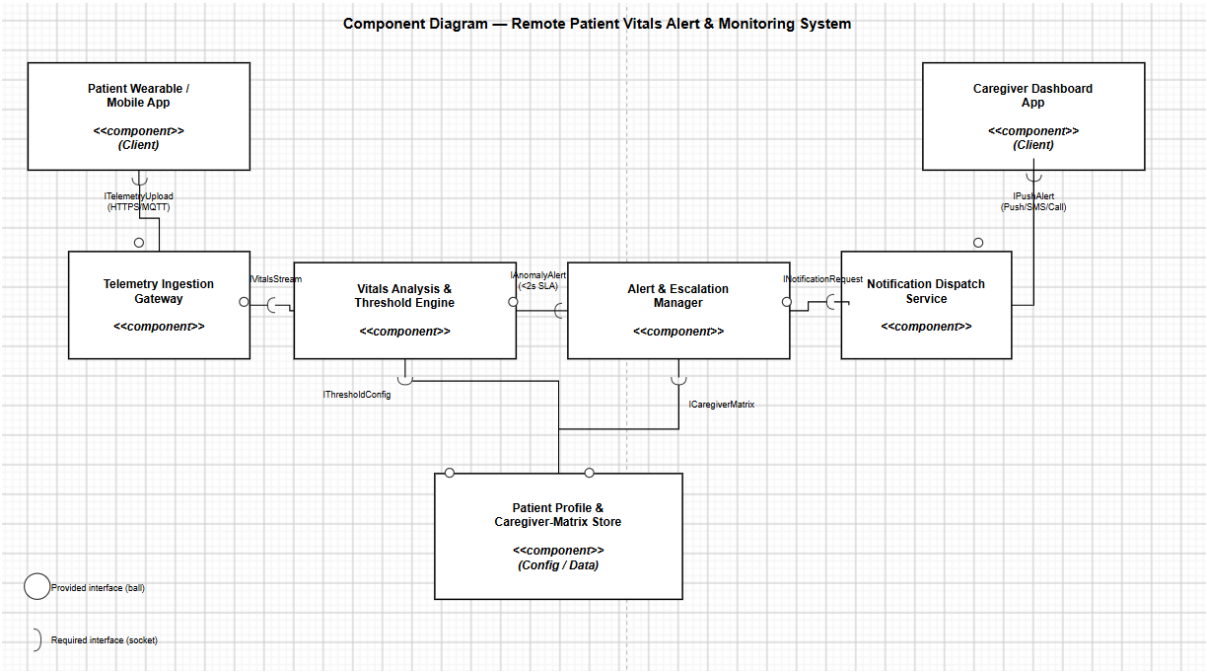
System Components

Five components for the Remote Patient Vital Alert & Monitoring System:

1. Patient & Caregiver User Interface
    - Displays patient vital signs such as SpO2, heart rate, and blood pressure
    - Displays alerts and emergency notifications
    - Allows caregivers to view patient vital history
    - Accepts patient and caregiver requests
2. Vitals Data Ingestion & Processing
    - Collects vital-sign data from remote monitoring devices
    - Validates and processes incoming readings
    - Detects device connectivity issues
    - Sends processed vital data for further analysis
3. Threshold Detection & Monitoring
    - Compares vital readings against predefined clinical thresholds
    - Identifies abnormal or critical readings
    - Maintains patient-specific threshold settings
    - Provides detected threshold breaches to the alert system

4. Alert & Emergency Escalation Manager
    - Generates real-time clinical alerts
    - Sends notifications to caregivers when abnormal readings are detected
    - Tracks alert acknowledgement
    - Escalates unacknowledged critical alerts to the next caregiver in the escalation matrix
5. Patient Health Database
    - Stores patient and caregiver information
    - Stores vital-sign readings and historical data
    - Stores threshold configurations
    - Stores alert and escalation records
    - Maintains device connectivity and monitoring information

MAIN COMPONENT DIAGRAM



Architectural Justification

We chose **Layered Architecture** for the **Remote Patient Vital Alert & Monitoring System**.

Reason 1 – Separation of Concerns

The architecture separates **patient/caregiver interaction, vital data processing, threshold detection, alert and emergency escalation, and data storage** into distinct layers. This makes the system easier to understand, develop, test, and maintain.

Reason 2 – Suitability

The system performs well-defined operations such as **collecting patient vital signs, processing readings, detecting threshold breaches, generating alerts, and escalating critical situations to caregivers**. Layered architecture provides a straightforward structure for these operations without the additional complexity of a distributed architecture.

Security Advantage

Sensitive patient health information can be isolated in the **database layer** and accessed through controlled application components rather than exposing the database directly to patients or caregivers. This helps protect vital records and other confidential healthcare information.

Performance Advantage

The business components can **process and analyze vital-sign readings before presenting results to users**. This reduces unnecessary processing at the user-interface level and allows alerts and critical information to be delivered efficiently.